

Outage Investigation Report —

OIR-2025-002

Field	Value
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Report Date	2025-09-12
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Reviewing Manager	Priya Ramachandran, Plant Manager, NW-PowerPool
Region	NW-PowerPool
Site	NW-PowerPool-WIND-024
Asset	WIN-10131 (Siemens Gamesa, wind_turbine)
Outage ID	OUT-2025-0570
Classification	Forced outage — overheating
Distribution	NW-PowerPool Operations, Reliability Engineering, OEM Liaison, Vendor Management

1. Outage Summary

WIN-10131 tripped on gearbox over-temperature on 2025-08-27 at 20:43 PDT, returning to service on 2025-09-04 at 18:43 PDT after a duration of **190 hours**. Total lost generation: **4,632.9 MWh**. This is the second multi-day overheating event on WIN-10131 in FY2025; the first was the January event recorded in OIR-2025-005 (139 h, 3,389.3 MWh).

This unit has now experienced 8 forced outages in FY2025 totaling 17,239 MWh of lost generation. The cumulative pattern is the subject of MOC-2025-022 (Siemens Gamesa OEM threshold override) and RCA-2025-005 (procedural shortcut analysis).

2. Timeline of Events

Timestamp (PDT)	Event
2025-08-26 13:20	Routine SCADA shows main bearing temperature rising 0.4 °C/hr, atypical for ambient 28 °C and 8 MW load.
2025-08-27 02:10	Bearing temp crosses 78 °C "monitor" threshold per Siemens Gamesa OEM Manual Rev. 4.2 §7.3.
2025-08-27 11:55	Bearing temp crosses 82 °C. Cascadia internal Standard MS-2025-03 §5.1 requires shutdown ≥ 80 °C for sustained > 30 min; OEM Manual permits run to 88 °C. Operator continued operation under the OEM threshold per pre-MOC standing instruction.
2025-08-27 20:43	Gearbox over-temperature protection trips at 91 °C oil-sump. Forced outage declared.
2025-08-27 22:15	CPS dispatch requested. CPS unable to mobilize within 4 hours due to concurrent NW-PowerPool-WIND-014 substation work.
2025-08-28 00:30	EPA-2024-11 invoked. Apex Emergency Grid Solutions accepts dispatch. Apex subcontracts Siemens Gamesa OEM crew at 1.8× rate per Apex MSA Schedule B-3.
2025-08-29 — 2025-09-02	Gearbox oil drain, inspection, sump heat-exchanger replacement, oil-spectrometer analysis. Spectrometer indicated iron particulate at 18 ppm (limit 12).
2025-09-03	Vibration baseline retest. Two restart attempts failed on high-side bearing temperature. Third attempt sustained.
2025-09-04 18:43	Returned to service at de-rated 6.2 MW pending OEM warranty review of full gearbox replacement quote.

3. Telemetry Observations

- Gearbox sump oil temperature at trip: 91.4 °C (limit 90).
- Main bearing temperature at trip: 96.2 °C.
- Vibration RMS on main shaft: 7.8 mm/s (Cascadia MS-2025-03 limit 6.5; OEM 10.0).
- Iron particulate in oil: 18 ppm.
- The CSM-7 quarterly oil-spectrometer test scheduled for July 2025 (WO-2025-07212) was closed as "deferred to next scheduled outage" per technician notes — see CAR-2025-006 (CSM-7 procedural shortcut).

- Ambient conditions: 28–31 °C daytime, 4.2 m/s SW wind. No icing or storm-related stressor.

4. Restoration Activities

Vendor	Scope	Invoice (USD)
Apex Emergency Grid Solutions	Mobilization, parts staging, prime-contractor management	\$164,000
Siemens Gamesa OEM (via Apex 1.8× subcontract)	Sump heat-exchanger replacement, oil change, vibration analysis, warranty assessment	\$221,000
Total		\$385,000

CPS scheduled equivalent estimate (issued 2025-09-10 for retrospective benchmark): \$96,500. See EMR-2025-002 for invoice detail and VRA-2025-003 for the Apex MSA pricing schedule.

Siemens Gamesa OEM has issued quotation Q-SG-25-0844 for full gearbox replacement at **\$1.2M**. The recommendation is based on vibration RMS exceeding OEM Manual Rev. 4.2 threshold of 6.0 mm/s on the high-speed shaft (measured 6.4 mm/s on the second restart attempt). Cascadia internal engineering disputes the replacement recommendation under MOC-2025-022; refer to RAM-2025-004.

5. Preliminary Root Cause

Direct cause: degraded gearbox oil with elevated iron particulate, leading to inadequate heat transfer in the sump heat exchanger and runaway bearing temperature under summer ambient load.

Contributing causes:

1. Deferral of CSM-7 oil-spectrometer test (WO-2025-07212) — refer to CAR-2025-006.
2. Continued operation between 82 °C and 90 °C bearing temperature, permitted by OEM manual but flagged by Standard MS-2025-03; pending MOC-2025-022.
3. WIN-10131 had a prior overheating event in January 2025 (OIR-2025-005) without root-cause closure — the deferred Q1 work was not re-prioritized following the January event.

6. Recommendations

1. Close MOC-2025-022 with a documented Cascadia-specific operating threshold. Pending engineering review since March 2025 (9 months as of this report).
2. Reinstate the CSM-7 quarterly oil-spectrometer step at NW-PowerPool. Track under CAR-2025-006.
3. Reject the Siemens Gamesa \$1.2M gearbox replacement quote pending Cascadia engineering evaluation. Refer to RAM-2025-004 and RCA-2025-003.
4. Add WIN-10131 to the FY26 capital planning watchlist alongside WIN-10011 and WIN-10130 (RCA-2025-003).
5. Cross-reference RCA-2025-002 (Fall Overheating Sequence) for the broader pattern.

Signatures

Role	Name	Date
Author	Elena Vasquez, Reliability Engineer II	2025-09-12
Plant Manager	Priya Ramachandran	2025-09-13
OEM Liaison	J. Bertelsen	2025-09-14

Distribution: NW-PowerPool Operations · Reliability Engineering · OEM Liaison · Vendor Management · CFO Office

References: OIR-2025-005 · CAR-2025-006 · EMR-2025-002 · VRA-2025-003 · MOC-2025-022 · RAM-2025-004 · RCA-2025-002 · RCA-2025-003 · RCA-2025-005 · Siemens Gamesa OEM Manual Rev. 4.2 · Standard MS-2025-03 · CSM-7